Original paper

Comparative evaluation of selective culture and PCR for detection of *Fusobacterium necrophorum* and *Fusobacterium nucleatum* in throat specimens

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Summary

Culture on selective media (*Fusobacterium* selective agar supplemented with 4 mg/L vancomycin and 8 mg/L neomycin) was compared to PCR (*rpo*B and *haem* genes for *Fusobacterium necrophorum*, 16S rDNA fragment specific for *Fusobacterium nucleatum*) for direct detection of the species in throat specimens of 88 patients with sore throat and pharyngotonsilits.

Among the 49 culture-positive specimens, culture and PCR revealed identical positive results in 38 (43.1%) and eight (9.1%) patients for *F. nucleatum* and *F. necrophorum*, respectively, whilst in two more patients (2.3%), both pathogens were detected by both methods. Only a single *F. nu-*

cleatum culture-positive specimen was PCR-negative. In contrast, among the remaining 39 culture-negative specimens, five (5.7%) and two (2.3%) were positive by PCR for *F. nucleatum* and *F. necrophorum*, respectively. Sub-species identification revealed that all *F. necrophorum* isolates were identified as *F. necrophorum* ssp. *funduliforme*. Sensitivity, specificity, PPV and NPV of the 16S rDNA PCR (F. nucleatum detection) were 97.6, 89.4, 88.9 and 97.7%, respectively, whilst the respective values for the rpoB PCR (*F. necrophorum* detection) were 100.0, 97.4, 83.3 and 100.0%, as compared to culture.

In conclusion, PCR proved of higher diagnostic yield than culture and detected both species in culture-negative specimens. As similar results have been obtained for other anaerobic species, the golden standard anaerobic culture seems to be insufficient for detecting *Fusobacterium* spp. in the area of molecular microbiology and needs further re-evaluation.

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Introduction

Fusobacterium is a gram-negative non-motile and obligate anaerobic pathogen that is consisted of at least 12 species. *Fusobacterium nucleatum* and *Fusobacterium necrophorum* are their main species identified in clinical infections and the latter is consisting of two subspieces, *F. necrophorum* subs. *necrophorum* (usually identified in animals and sporadically in humans) and *F. necrophorum* subs. *funduliforme* (identified in humans).[1-3] *F. necrophorum* is the causative agent of Lemierre's disease, but is also detected in cases of pharyngotonsilitis and acute sore throat [3], while *F. nucleatum* is considered as pathogen in periodontal disease, but also in other human infections. Both species can be isolated in the microbiota of several parts of the human body (mainly oropharyngeal, urogenital and gastrointestinal), although their connection with the true normal flora is ambiguous.[4]

Detection of *Fusobacterium* species in clinical specimens is usually based on conventional culture methods, although molecular techniques have also been proposed as alternative identification assays.[3-6] Culture can be performed using non-selective but also selective media that allow growth of the pathogen and inhibition of the microbiota.[5] Nevertheless, the true clinical significance of a culture-positive and/or a PCR-positive specimen for *Fusobacteriun* spp. in the absence of concomitant symptoms and signs has not been elucidated. In addition, a direct comparison between PCR assays and selective culture has only been performed for *F. necrophorum* but not for *F. nucleatum*.[7] PCR was found to be more sensitive and of higher diagnostic yield than culture among other anaerobic species, but culture is still considered the golden standard of anaerobic bacteriology.[8]

The aim of the present study is a comparative evaluation of culture on selective media and PCR assays for the identification of *F. nucleatum* and *F. necrophorum* in clinical specimens of patients with sore throat and pharyngotonsilits.

Materials and methods

Setting and sample collection: During the period January-April 2014, throat specimens were collected from patients presented in two outpatient hospital departments in Athens, Greece and examined for sore throat and/or pharyngotonsilitis.

The throat swabs were re-suspended in 500 μl of sterile 0.9% NaCl and divided in three parts. Part A (100 μl) and part B (200 μl) were used for culture and DNA extraction, respectively. Part C (200 μl) was kept in deep refrigeration (-80° C).

Non-selective and selective culture: General anaerobic culture was performed from part A of the specimen on CDC-anaerobe agar plates (Bioprepare, 19001, Keratea, Greece) for evaluation of the microbiota. Culture for *Fusobacterium* spp. was performed from part A of the specimen on Fusobacterium selective agar plates supplemented with 4 mg/L vancomycin and 8 mg/L neomycin (Bioprepare).[5] Culture for *Streptococcus pyogenes* was performed on 5% sheep blood agar plates (Bioprepare). All plates were incubated under anaerobic conditions for 72 hours.

Identification of *Fusobacterium* species was performed using standard methodology, supplemented with the BBL Crystal Anaerobe system (Becton Dickinson and Co, 07417-1880, USA).[5] Identification of *S. pyogenes* was performed using standard methodology, supplemented with the the latex agglutination test for the Lancefield group A antigen (Lorne Laboratories Ltd, Reading, UK), and the API 20 STREP identification system (bioMerieux, Marcy L'Etoile, France).

PCR assays: DNA was extracted from part B of the specimen using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) and the tissue protocol, according to the manufacturers' instructions. PCR reactions, targeting the *rpo*B gene of *F. necrophorum* (present in both *F. necrophorum* subspecies), the *haem* gene specific for *F. necrophorum* ssp. *necrophorum* and the 16S rDNA of *F. nucleatum*, were performed using previously published protocols and procedures.[3,6]

Sensitivities, specificities, positive predictive (PPV) and negative predictive values (NPV) of the PCR assays were calculated based on the comparison with the culture as the cold standard.

Clinical and epidemiological data: The following data were obtained from all patients: gender, age, fever, lymphadenopathy, other infection, other predisposing factors (asthma, cardiovascular disease, diabetes, and autoimmune disease). Patients that received antimicrobial treatment of any kind during the preceding thirty days were excluded from the study.

Statistical analysis: Statistical analysis was performed using the chi square and/or the Fisher's exact tests as appropriate. All *p* values reported are two-tailed and statistical significance was set at 0.05.

Results

A total of 88 patients were enrolled in the study, of which 57 (65%) were male and 31 (35%) female and an equal number of pharyngeal specimens were obtained.

Culture and PCR results of these specimens are shown in Table 1. Briefly, 49 specimens (55.7%) were culture-positive for *Fusobacterium* spp. and 39 were culture-negative. Of the 49 culture-positive specimens, *F. nucleatum* was isolated in 39 (44.2%), *F. necrophorum* in eight (9.1%), whilst in two specimens (2.3%) both species were detected.

Of the 39 specimens that were found by culture to harbor only *F. nucleatum*, 38 were also PCR positive for the 16S rDNA F. nucleatum-specific fragment and one

Table 1 Cumulative culture and PCR results of the 88 specimens

Culture results		PCR results		Patients (n = 88)	
F. nucleatum	F. necrophorum	F. nucleatum	F. necrophorum	No	(%)
Positive	Negative	Positive	Negative	38	(43.2)
Positive	Negative	Negative	Negative	1	(1.1)
Negative	Positive	Negative	Positive	8	(9.1)
Positive	Positive	Positive	Positive	2	(2.3)
Negative	Negative	Positive	Negative	5	(5.7)
Negative	Negative	Negative	Positive	2	(2.3)
Negative	Negative	Negative	Negative	32	(36.3)

was negative, whilst all 39 were negative for the *rpo*B *F. necrophorum*-specific fragment. The 8 specimens that were found by culture to harbor only F. necrophorum were also all PCR positive for the rpoB fragment. Finally, the two specimens that were found to harbor both species by culture were also PCR-positive for 16S rDNA and *rpo*B fragments. *F. necrophorum* subspecies identification revealed that all ten were identified as *F. necrophorum* ssp. *funduliforme*.

Of the 39 culture-negative specimens, 32 were PCR-negative for both the 16S rDNA and the *rpo*B fragments, five were *F. nucleatum* PCR-positive (16S rDNA fragment), whilst two were *F. necrophorum* PCR-positive (*rpo*B fragment), which were further identified as *F. necrophorum* ssp. *funduliforme*.

Furthermore, in 21 specimens, culture yielded also *S. pyogenes* but without any statistically significant association with the *Fusobacterium* spp. culture or PCR results.

Sensitivity, specificity, PPV and NPV of the 16S rDNA PCR (*F. nucleatum* detection) were 97.6, 89.4, 88.9 and 97.7%, respectively, whilst the respective values for the *rpo*B PCR (*F. necrophorum* detection) were 100.0, 97.4, 83.3 and 100.0%, as compared to the gold standard which was the culture on selective media.

Regarding the clinical and epidemiological investigation, mean age was 28.1 years (standard deviation 11.9, range 14-78 years) and median age was 24 years. All patients presented with sore throat. Local lymphadenopathy, fever, other infection with similar symptoms during the previous month, or other underlying disease were present in 52 (59.1%), 49 (55.7%), 20 (22.7%) and 4 (4.5%) patients, respectively.

Statistical evaluation of the culture and PCR results, together with epidemiological and clinical data revealed a statistically significant association only between culture and PCR results for detection of *F. nucleatum* or *F. necrophorum* (both $p<0.01$). No other statistically significant associations were detected.

Discussion

Species of the *Fusobacterium* genus are considered to be part of the upper respiratory tract microbiota, although they have also implicated in various upper respiratory infections. In that respect, clinical laboratories only occasionally detect the genus in everyday workflow, as the procedure of anaerobic culture is laborious and time consuming. Alternatively, PCR techniques have been used to facilitate better results, given the fast turnaround times and high sensitivity.[3,6] Nevertheless, culture is still considered the gold standard for anaerobic infection diagnosis as well as for evaluation of novel techniques in comparative studies. Surveys for detection of Fusobacteria using selective media are scarce in the literature and apply only to *F. necrophorum* and not to *F. nucleatum*.[7,9] Direct comparative studies of selective culture methodologies with PCR are also limited in the literature.[7]

In that respect, in the present study a direct comparison of PCR and culture for detection of both main fusobacteria species in throat specimens was performed using a selective medium containing vancomycin (4 mg/L) and neomycin (8 mg/L). In the previous comparative study selective media containing vancomycin (2.5 mg/L) and nalidixic acid (5 mg/L) were used.[7] Fusobacteria are vancomycin-resistant and vancomycin supplement is the corner stone of all selective media. Direct comparison of the two media was not performed, as the medium used in the present study performed well and no mixed cultures were detected (data not shown), as suggested also by the Wadsworth Anaerobic Manual.[5]

Our study indicated that among culture-positive specimens, the association between culture on selective media and PCR was high and the results were almost identical, as only a single specimen produced discrepant results (culture-positive for *F. nucleatum*, but PCR-negative). In contrast, among the 39 culture-negative specimens, a considerable number was PCR-positive for one of the two pathogens (five for *F. nucleatum* and two for *F. necrophorum* ssp. *funduliforme*), thus indicating that PCR may detect these pathogens where culture fails.

Similar results regarding the higher yield of PCR in comparison to culture were obtained also by Bank et al[7], which however was focused only on *F. necrophorum*. Nevertheless, our study indicated a lower detection rate of 11.4% (10/88) for *F. necrophorum* as compared to 30.2% (45/139) in that study, and a high detection rate of 46.6% (41/88) for *F. nucleatum* (but no comparison with other studies is available).[7] This latter can be explained due to the implication of *F. nucleatum* in periodontal disease, but also its frequent detection among the oropharyngeal microbiota in healthy volunteers.[6]

In addition, another recent study regarding a different anaerobe, *Bacteroides fragilis* [8], indicated also that a number of culture-negative specimens harbored the pathogen (as shown by two different PCR targeting different DNA fragments), thus confirming the higher yield of the PCR assay. This study suggested also a modification of the golden standard procedure, thus allowing incorporation of the PCR, together with culture in the golden standard evaluation procedure. It should be noted that the use of two different targets in the previous study proved beyond any doubt that

PCR contamination did not occur and all PCR-positive were true positive.[8]

In conclusion, the present study confirms previous results regarding *F. necrophorum*, and indicates for the first time, that PCR assays have higher diagnostic yield than selective culture for detecting *F. nucleatum*. In that respect, PCR should be incorporated in the standard procedure, as it has also suggested for other anaerobic pathogens.[8] Nevertheless, the true clinical significance of detecting *Fusobacterium* species in throat swabs specimens, and the association with various clinical presentations was not evaluated as it was not within the scope of this study, and further studies are needed to elucidate it.

Conflicts of interest
None to declare by any author

Περίληψη

Συγκριτική αξιολόγηση της καλλιέργειας σε εκλεκτικά υλικά και της PCR για την ανίχνευση του *Fusobacterium necrophorum* και του *Fusobacterium nucleatum* σε φαρυγγικά δείγματα

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Στην παρούσα μελέτη διενεργήθηκε σύγκριση της καλλιέργειας σε εκλεκτικά υλικά (Fusobacterium άγαρ εμπλουτισμένο με 4 mg/L βανκομυκίνη και 8 mg/L νεομυκίνη) με την PCR (με στόχους τα γονίδια *rpo*B και *haem* του *Fusobacterium necrophorum* και τμήμα του 16S rDNA ειδικό του *Fusobacterium nucleatum*) με σκοπό την άμεση ανίχνευση των ειδών σε φαρυγγικά δείγματα 88 ασθενών με εικόνα φαρυγγοαμυγδαλίτιδας.
Συνολικά δείγματα από 49 ασθενείς ήταν θετικά με την καλλιέργεια, στα οποία η PCR έδωσε θετικά αποτελέσματα σε 38 (43,1%) και οκτώ (9,1%) για τα είδη *F. nucleatum* ή *F. necrophorum*, αντίστοιχα, ενώ σε δύο ακόμη ασθενείς (2.3%) αμφότερα τα παθογόνα ανευρέθηκαν. Μόνο ένα δείγμα, θετικό με την καλλιέργεια για το είδος *F. nucleatum* ήταν αρνητικό με την PCR. Σε αντίθεση, στα δείγματα των υπόλοιπων 39 ασθενών, τα οποία ήταν αρνητικά με την καλλιέργεια, πέντε (5,7%) και δύο (2,3%) ήταν θετικά με την PCR για τα είδη *F. nucleatum* ή *F. necrophorum*, αντίστοιχα. Η ταυτοποίηση σε επίπεδο υπο-είδους ανέδειξε ότι σε όλα τα δείγματα που ανευρέθηκε το είδος *F. necrophorum*, αυτό ταυτοποιήθηκε ως *F. necrophorum* ssp. *funduliforme*. Η ευαισθησία, ειδικότητα, θετική και αρνητική προγνωστική αξία της PCR για το 16S rDNA (ανίχνευση του *F. nucleatum*) ήταν 97,6%, 89,4%, 88,9 και 97,7%, αντίστοιχα, ενώ οι αντίστοιχες τιμές για την PCR του γονιδίου *rpo*B (ανίχνευση του *F. necrophorum*) ήταν 100,0%, 97,4%, 83,3 και 100,0%, σε σύγκριση με την καλλιέργεια.
Συμπερασματικά, η PCR αποδείχθηκε ότι διαθέτει υψηλότερη δυνατότητα ανίχνευσης από την συμβατική καλλιέργεια σε εκλεκτικά υλικά, αφού ανίχνευσε τα παθογόνα σε κλινικά δείγματα αρ-

νητικά με την καλλιέργεια. Λαμβάνοντας υπ' όψιν ότι παρόμοια αποτελέσματα έχουν καταγραφεί και σε άλλα αναερόβια είδη, το «χρυσό πρότυπο» της αναερόβιας καλλιέργειας χρειάζεται επαναξιολόγηση στην εποχή της μοριακής μικροβιολογίας.

Λέξεις κλειδιά
PCR, καλλιέργεια, Fusobacterium nucleatum, Fusobacterium necrophorum, φαρυγγοαμυγδαλίτιδα

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